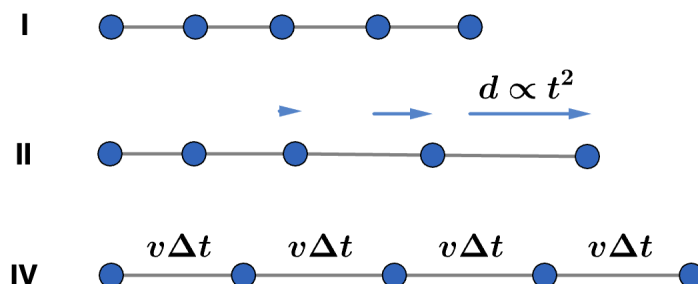


2017 F=ma Exam: Problem 4

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Initially, the cars are equally spaced with negligible distance between them as shown in pattern I.



When the cars start accelerating, the distance they travel is

$$d = \frac{1}{2}at^2 \propto t^2$$

Since $d \propto t^2$,

$$\Delta d = d_n - d_{n-1} \propto n^2 - (n-1)^2 = 2n - 1$$

so the distance between adjacent cars increases the longer they have been traveling. Thus, car spacing increases from left to the right as shown in pattern II.

Once the cars all reach the speed limit v , the distance between adjacent cars stays constant. Since each car spends Δt (reaction time) more time going at v than the car behind it, they are all equally separated (by $v\Delta t$) as shown in pattern IV.

The car pattern evolves from I to II to IV so the answer is B.